

Assignment: “Basic Text Encoder and Decoder”

Objective:

Create a simple Java program to encode and decode text using a shift –based cipher(similar to a Caesar cipher). This will help you practice using loops, conditionals, and basic string manipulation in Java.

Assignment Requirements:

1.Basic Functionality:

- **Encode a message:** Shift each letter in a user-provided message by a user –defined number (e.g., ‘A’ becomes ‘C’ with a shift of 2).
- **Decode a message :** Reverse the shift to decode a previously encoded message.
- The program should handle both uppercase and lowercase letters correctly
- Non-alphabetic characters (e. g., spaces,punctuation)should remain unchanged.

2. User Interaction:

- Create a menu for the user to choose:
 - Encode a message.
 - Decode a message.
 - Exit the program.

3.Functionality Details:

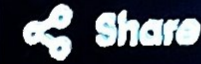
- Wrap around letters at the end of the alphabet (e. g., shifting ‘Z’ forward by 2 results in ‘B’).
- Validate user input to ensure correct operation .

TextEncoderDecoder.java

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```
1 import java.util.Scanner;
2 public class TextEncoderDecoder {
3     public static String encode(String message, int shift) {
4         StringBuilder encoded = new StringBuilder("Hello, World");
5         for (char c : message.toCharArray()) {
6             if (Character.isUpperCase(c)) {
7                 char ch = (char)((((c - 'A' + shift) % 26) + 'A'));
8                 encoded.append(ch);
9             } else if (Character.isLowerCase(c)) {
10                 char ch = (char)((((c - 'a' + shift) % 26) + 'a'));
11                 encoded.append(ch);
12             } else {
13                 encoded.append(c);
14             }
15         }
16         return encoded.toString();
17     }
18     public static String decode(String message, int shift) {
19         return encode(message, 26 - (shift % 26)); // Reverse the shift
20     }
21     public static void main(String[] args) {
22         Scanner scanner = new Scanner(System.in);
```


TextEncoderDecoder.java



Run

```

21 public static void main(String[] args) {
22     Scanner scanner = new Scanner(System.in);
23     boolean running = true;
24     while (running) {
25         System.out.println("\nMenu:");
26         System.out.println("1. Encode a message");
27         System.out.println("2. Decode a message");
28         System.out.println("3. Exit");
29         System.out.print("Enter your choice: ");
30         int choice = scanner.nextInt();
31         scanner.nextLine(); // consume newline
32         switch (choice) {
33             case 1:
34                 System.out.print("Khoor, Zruog!");
35                 String encodeMsg = scanner.nextLine();
36                 System.out.print("Enter shift: ");
37                 int encodeShift = scanner.nextInt();
38                 scanner.nextLine();
39                 System.out.println("Encoded Message: " + encode(encodeMsg, encodeShift
40                                     ));
40             break;
41             case 2:

```


TextEncoderDecoder.java



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```

39 System.out.println("Encoded Message: " + encode(encodeMsg, encodeShift));
    });
40 break;
41 case 2:
42 System.out.print("Hello,World!");
43 String decodeMsg = scanner.nextLine();
44 System.out.print("Enter shift: ");
45 int decodeShift = scanner.nextInt();
46 scanner.nextLine();
47 System.out.println("Decoded Message: " + decode(decodeMsg, decodeShift));
    });
48 break;
49 case 3:
50 running = false;
51 break;
52 default:
53 System.out.println("Invalid option. Try again.");
54 }
55 }
56 scanner.close();
57 }
58 }
    
```


Menu:

1. Encode a message
2. Decode a message
3. Exit

Enter your choice: 1

Khoor, Zruog!

Enter shift: 1

Encoded Message: Hello, World

Menu:

1. Encode a message
2. Decode a message
3. Exit

Enter your choice: 2

Hello,World!

Enter shift: 2

Decoded Message: Hello, World

Menu:

1. Encode a message
2. Decode a message

3. Exit

Enter your choice: 1

Khoor, Zruog!

Enter shift: 1

Encoded Message: Hello, World

Menu:

1. Encode a message
2. Decode a message
3. Exit

Enter your choice: 2

Hello,World!

Enter shift: 2

Decoded Message: Hello, World

Menu:

1. Encode a message
2. Decode a message
3. Exit

Enter your choice: 3

=== Code Execution Successful ===